Abstract

PROBLEM :

Urban areas face two persistent challenges — inefficient waste management and rising energy demands. Overflowing bins cause pollution and hygiene issues and waste in landfills releases methane, and leads to land and air pollution.

SOLUTION:

The SmartWaste-to-WiFi System transforms garbage into green energy. Smart bins equipped with IoT sensors monitor waste levels and prevent overflow of garbage and makes the community clean .Collected organic waste is processed in anaerobic digesters, where it generates biogas used to power public Wi-Fi . Each digester is coated with a photocatalytic algae to enhance gas production and purify surrounding air.

METHODOLOGY:

IntegratedIOTbasedwaste energy conversion system .

BENEFITS:

Converts organic waste into renewable energy.
Reduces landfill load and greenhouse gas emissions. Provides sustainable power for public connectivity. Enhances air quality through photocatalytic algae coating. Supports economy and smart city development

SDGs: 7,11,12,13

Images of Proposed Prototype/Innovation

(Sampleexpectedoutput / result)

Define the problem and its relevance to today's market

1. Ineffective waste management, limited public Wi-Fi access, and rising air pollution hinder sustainable development and human well-being.

2. Poorwaste disposal practices threaten health and the environment, Additionally, increasing air pollution endangers public health and accelerates environmental degradation . 3. Addressing these issues requires integrated solutions such as community-driven waste management, innovative recycling and waste- to-energy technologies, expansion of public Wi-Fi ,and affordable air purification systems. 4. Together, these promote environmental resilience and sustainabale development.

Describe the Proposed Solution

Smart Reward-Based Bin System

1. Smartbins use sensors,IoT,QRcode to identify waste types, track data, and efficient waste monitoring.
2. Rewardmechanism motivates citizenstosegregatewaste properly,strengthens recycling habits, supports local businesses, and increases community participation actively.

Anaerobic Digestors

1. Anaerobic digestion converts organic waste into methane-rich biogas, reducing landfill waste while producing reliable renewable energy locally.
2. Generated biogas powers streetlights and public Wi-Fi hubs, lowering carbon emissions and making essential urban services energy independent.

Algae-Based Photocatalytic Coating

1. Digestate combined with microalgae and TiO_2 ZnO nanoparticles forms a photocatalytic coating with environmental purification properties.
2. This living coating absorbs carbon dioxide, degrades air pollutants, enhances biogas yield, and improves overall urban air quality

Describe Your Technology Stack
(Hardware & Software details)

HARDWARE :

- Ultrasonic Sensors
- Load cell
- Arduino uno, ESP 32
- QR Code
- Servo DC Motors
- moisture sensor
- IR sensor
- DC geared motor
- Shaft coupler
- Solar panel
- Sensing plate
- Lcd display
- Anaerobic Digester Tank
- Pressure Gauges
- Biogas engine
- Generator
- PH & thermometer

SOFTWARE:

IOTclouddashboard Arduino IDE
HX711 library
firebase
MQTT/REST
PYTHON , C++,HTML, MANGODB

ESP 32

SERVO MOTOR

ULTRASONIC SENSOR

IR SENSOR

How your proposed solution is different from similar kind of product by the competitors (ie. Market) if any

SMART BIN :

Existing: Companies like BIN-E already make bins which rewards only for plastic waste and dont involve bio wastes . Their goal is to just reduce plastic waste.

Difference: Our binsdon’tjust detect waste they rewardpeople fordoingthe right thing.Every user is identified by a QR code when they dispose of bio-waste correctly, they earn reward points through a website.Thus it improves user contribution

(we focus on people + motivation + community benefit.)

ANAEROBIC DIGESTOR TANKS:

Existing: Some companies use large biogas plants which are centralized and costs money and fuel.
Difference: We use compact digesters installed near communities.It generates biogas which is later converted to electricity to power wifi hubs and the leftover waste becomes organic fertilizer.

ALGAE COATING :

Existing: Photocatalytic panels exist but they're all separate technologies.
Difference: We coat the algae mixture on the digester tank to purify the surrounding air and as well as air from the tank

Explain the uniqueness and distinctive features of the (product / process / service) solution

1. This project is unique because it links waste management, renewable energy, and environmental purification into one smart system.
2. Unlike existing bin that only focus on waste collection or recycling, our solution combines three technologies to maximize impact and efficiency.
3. Anaerobic digester converts biodegradable portion to renewable energy.
4. Alage coating enhances energy output while purifying the air.
5. Together these features make the system distinctive by merging technology, sustainability, and community participation.
6. It’s not just a smart bin it’s a complete eco-innovation that rewards people, powers communities, and purifies the environment.

Utility: Highlight the utility/value proposition (key benefits) aspects of the solution/innovation

- This innovation creates a complete waste-to-value ecosystem that transforms everyday waste into energy, rewards, and cleaner air & environment and economic, social benefits all at once.

- The Smart Reward-Based Bin System makes people part of the solution and helps in proper diposal of waste .
- It improves community participation, ensures cleaner surroundings, and encourages recycling at the source.
- The Biodegradable Waste-to-Energy Unit converts organic waste into biogas, providing renewable energy for powering public spaces
- The Algae Coating does 3 jobs at once absorbs co2, purifies air and boots methane yield

- These offers a real, practical value that is cleaner cities, empowered communities, and a circular system where waste truly becomes a resource.

Scalability: Highlight the market potential aspects of the Solution/Innovation (Potential Market Size, Segmentation and Target users/ Customers)

With rapid urbanization and increasing waste generation, India alone produces over 62 million tonnes of solid waste annually, of which more than 70% remains untreated. This innovation fits perfectly into that market by combining IoT, renewable energy, and sustainability.

MARKET SIZE :

The Indian wastemanagement market was valued at approximately USD 14 billion in 2023 and is projected to reach USD 35 billion by 2030. Globally, the smart waste management industry is expected to surpass USD 5.5 billion by 2030, ensuring strong international potential as well.

SEGMENTATION :

- MunicipalCorporations seeking efficient waste management solutions.
- Educational Institutions implementing sustainability initiatives.
- Local Businesses participating in the reward system.
- Residents & Students who actively engage through reward system

GOVE RNMENT OBJECTIVES PROJECT ACHIEVES

- Swachh bharat mission
- Smart cities mission
- National Bio-Energy Mission
- National Clean Air Programme

SOURCES OF FUNDING

Our project is not dependent on a single funding source.

- government support,
- CSR funding,
- impact investors, and
- Institutions

Economic Sustainability: Highlight Commercialization / Business application aspects (how it is going to economic profitable and viable)

- **Energy sales**: The biogas generated from the anaerobic digesters is a renewable energy source that can be sold and digestate can be sold to local farms .
- **Carbon credit trading**: By reducing methane emissions from landfills and generating renewable energy, the project can earn and sell carbon credits in the growing carbon market.

- **Data monetization**: The IoT sensors in the smart bins collect data on waste generation patterns, fill levels, and community usage of the Wi-Fi. This data can be analyzed and sold to market research firms, urban planners and etc.

- **Business Collaboration**: Through the reward system when users earn points for proper waste disposal, they redeem them at these partnered outlets. This creates a small business which is a mutually beneficial .
- By diverting organic waste from landfills, the system reduces tipping fees associated with landfill disposal

Environmental Sustainability and related benefit of the solution

- This project integrates smart waste segregation, renewable energy generation, and air purification into one sustainable system. The
- Smart Reward-Based Bin reduces landfill waste by encouraging source-level segregation.
- Organic waste is converted into biogas for clean energy as alternative for fossil fuels, while the digestate is reused as organic fertilizer. Algae-
- based coatings absorb CO_2 and reduce air pollution. The project supports a low-carbon
- future and aligns with UN SDGs 7, 11, 12, and 13.

Feasibility of the Solution/Innovation

- Each component of the system integrates existing, validated technologies
- Smart Reward-Based Bin System employs low-cost IoT modules .
- The Biodegradable Waste-to-Energy Unit is based on the well-established anaerobic digestion process, using compact biogas digesters. The technology is mature, safe, and adaptable, ensuring continuous biogas and compost production. The system requires minimal manual intervention due to automation and real-time monitoring.
- Partnerships with recyclers, local shops, and energy agencies enhance implementation
- Overall, the proposed model is technically, operationally practical, and economically sustainable, making it highly feasible for real-world deployment.

NUMERICAL DATAS

- **1kg** of food waste = **0.08-0.12** m^3 of biogas
  So for **50kg = 4-6** cub.metre of biogas

- **1**m^3 of biogas = **2kwh** of electrical energy (30-35%)
  So for **5**m^3 = **8-10**kwh of electricity

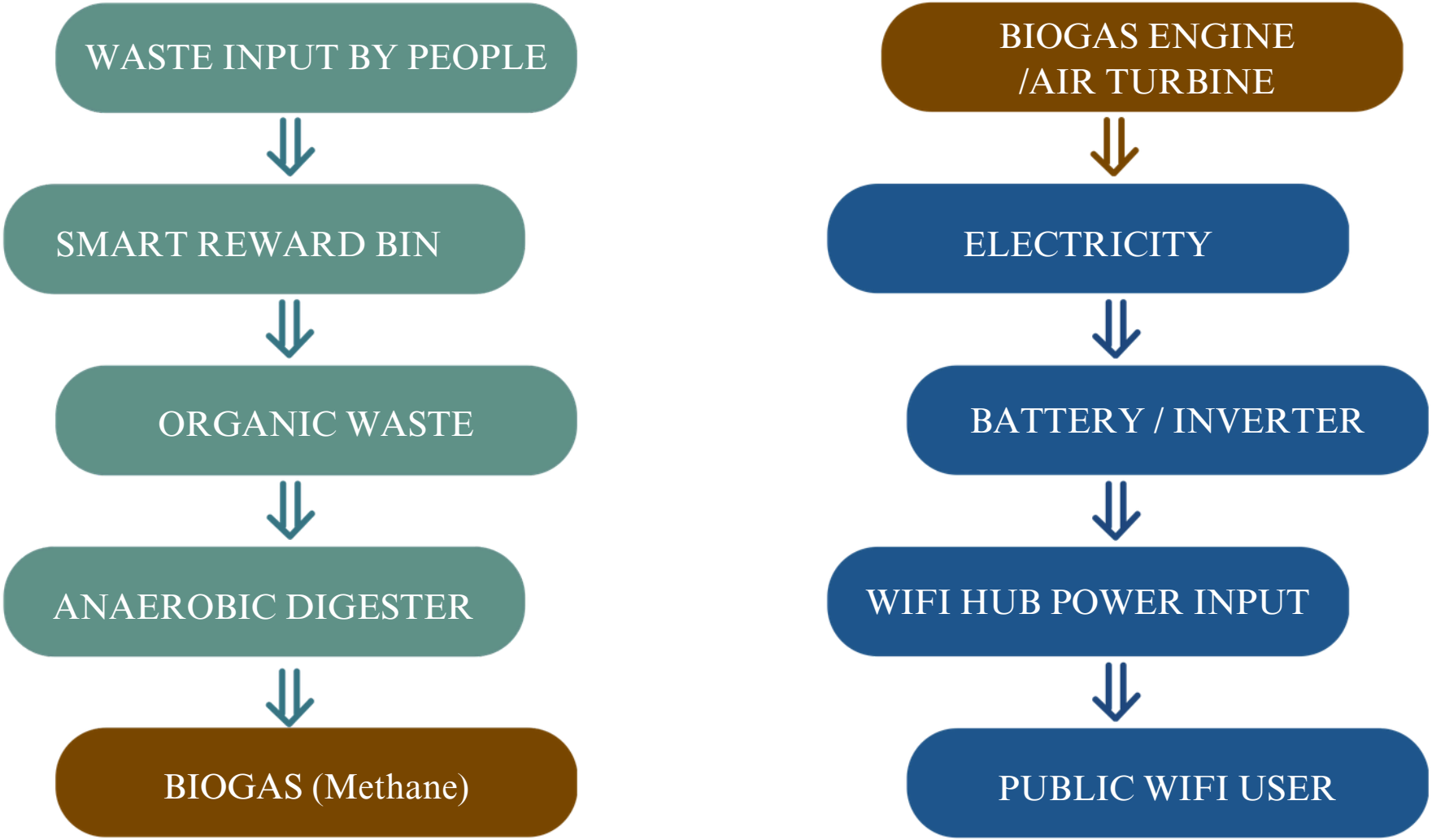
<u>10 kwh power can be used for:</u>

1.router - (24 - 48 hours)
2. 10 LED street light glow for (5-6 hour)
3. smart bin + sensor - for many days

Effective purification radius of algae coating =**3-6/meters** around coated surface

Estimated cost of the model : Rs. 30,000

FLOW CHART


WASTE INPUT BY PEOPLE
SMART REWARD BIN
ORGANIC WASTE
ANAEROBIC DIGESTER
BIOGAS (Methane)
BIOGAS ENGINE /AIR TURBINE
ELECTRICITY
BATTERY / INVERTER
WIFI HUB POWER INPUT
PUBLIC WIFI USER